

PHILIPS AKINBAMI

College Park, MD • phil.akinbami@gmail.com • (443) 946-9616 • LinkedIn • GitHub • Portfolio

EDUCATION

University of Maryland

M.S. in Telecommunications

College Park, MD

May 2026

Relevant Coursework: Introduction to Digital Communications Systems, Networks and Protocols I & II, AI Applications for 6G, Cellular Communication Networks, 5G System Design and Evolution.

Howard University

B.S. in Electrical Engineering, GPA 3.70

Washington, DC

May 2024

Relevant Coursework: Wireless Communication, Communications Theory, Electromagnetics, Signals and Systems, Electronics & Solid-State Devices, Introduction to Computer Networks.

Honors: \$15,000 NSBE Scholarship (academic excellence & leadership) · Dean's List, Fall 2023 & Spring 2024.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MATLAB, Pandas, NumPy, Matplotlib, Excel, Power BI, Git

Networking & Protocols: TCP/IP, IPv4/IPv6, VLANs, STP, SDN, DHCP, DNS, LLDP, SNMP, Routing & Switching

Automation & Analysis Tools: Ansible, NETCONF, RESTCONF, RESTful APIs, Wireshark, tcpdump, GNS3, Juniper Junos

Systems, Cloud & Infrastructure: Linux, Bash/Shell, VMware, Docker, AWS

Certification (in progress): JNCIA-DevOps

PROJECTS

MAWI Network Traffic Analyzer | Python, Pandas, Streamlit, PyShark, Wireshark, TCP/IP

[GitHub](#)

- Parsed **904,000+** MAWI backbone packets and engineered **17 time-windowed features** covering traffic volume, protocols, endpoint activity, and packet-size behavior.
- Identified **6,403 TCP retransmissions** and calculated reliability metrics for large-scale traffic analysis and troubleshooting.
- Built an interactive Streamlit platform with KPI summaries, filters, protocol views, retransmission trends, and downloadable feature data.

5G Integration and Test Lab | Open5GS, srsRAN, Docker, Ansible, Python, Linux, Bash

[GitHub](#)

- Building a **virtual 5G Standalone testbed** that integrates an srsRAN gNB and virtual UE with a containerized Open5GS core to support registration, PDU-session, connectivity, and performance testing.
- Built Open5GS from source using a **multi-stage Docker workflow** and deployed **NRF, UDR, UDM, AUSF, and AMF** services with MongoDB persistence, SBI networking, and network-function discovery.
- Configured and validated **PLMN, TAC, network slice, NGAP/SCTP, heartbeat, and network-function registration**; continuing development of SMF/UPF, ZeroMQ radio integration, iperf3 validation, and automated failure-recovery testing.

Cell Site Energy Analyzer | Python, pandas, Matplotlib

[GitHub](#)

- Processed **2.3M Telecom Italia cellular activity records** across **10,000 Milan grid cells** to identify a high-demand area and derive a normalized 24-hour traffic profile for detailed analysis.
- Developed a configurable traffic-to-power and battery model that estimated **47.69 kWh** of daily site energy consumption and backup runtime for a modeled **10 kWh** battery system.
- Identified an **12-hour risk window** in which estimated runtime fell below the 4-hour resilience target, reaching **2.4 hours** at peak traffic, and generated automated CSV reports and Matplotlib visualizations.

WORK EXPERIENCE

Cummins | System Analytics and Modeling Intern

Columbus, IN · May 2023 – Aug 2023

- Created Power BI dashboards used by **20+ engineers** to monitor KPIs, improve visibility into operational performance, and support data-driven decision making.
- Processed and consolidated datasets of **~100,000+ records**, improving reporting accuracy and enabling reliable KPI tracking for engineering teams.
- Authored step-by-step documentation supporting the team's migration from MATLAB workflows to Power BI, improving data-visualization consistency.

Cummins | Reliability Engineering Intern

Columbus, IN · May 2022 – Aug 2022

- Validated components using MATLAB and Excel-based reliability models across **10+ design evaluation cycles**.
- Documented failure trends and system performance issues, producing **8+ technical reports** for engineering review.
- Designed an onboarding database that streamlined training and improved workflow efficiency for a team of **15+ members**.